Class	Adm No

Candidate Name: ______________________________

millennia institute

Preliminary Examinations II 2009
Pre-university 3

H2 CHEMISTRY **9746 / 01**

Friday 18 Sep 2009 1h

Additional materials:
OMR
Data Booklet

READ THESE INSTRUCTIONS FIRST

1. Do not turn over this question paper until you are told to do so.
2. Write your name, class and index number in the spaces provided at the top of this page and on any separate answer paper used.
3. Attempt **ALL** the questions in Section A and B.
4. All answers are to be shaded on the OMR form provided using a 2B pencil.
6. Hand in your answers to the paper (OMR form) and question paper separately.

INFORMATION FOR CANDIDATES

No extra time will be given for shading of OMR form.

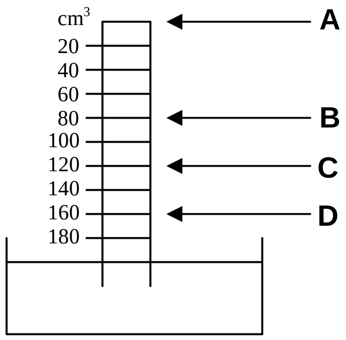
SECTION A (30 MARKS)
Answer all questions on the OMR form provided

For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct.

1. A tube is filled with 40 cm^3 of methane and 160 cm^3 of oxygen at room temperature. The open end of the tube is placed in a beaker of KOH(aq) as shown. The gas mixture was sparked to make the following reaction take place:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

What will be the final level of liquid in the tube after it has cooled back to room temperature?



2. Ions of two isotopes of the transition metal nickel are shown below.

$${}^{58}_{28}Ni^{2+} \qquad {}^{60}_{28}Ni^{2+}$$

Which one of the following statements is correct?

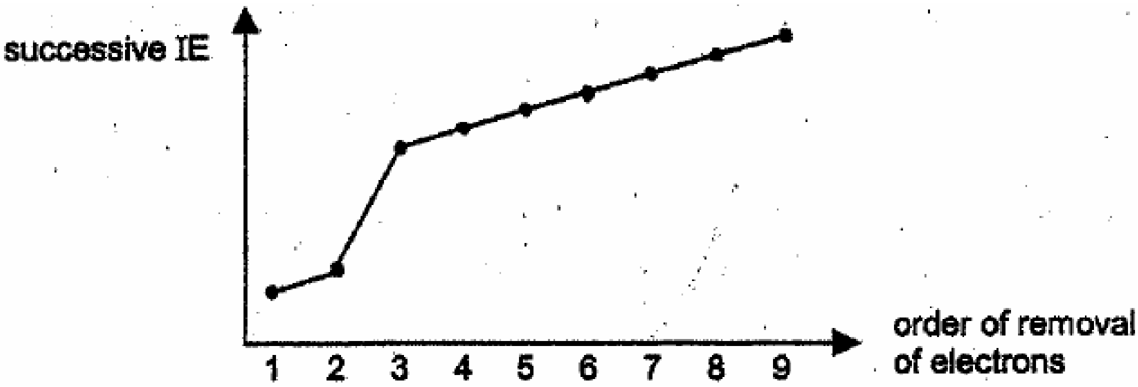
A The electron arrangement of both these Ni^{2+} ions is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$.

B The ${}^{60}_{28}Ni^{2+}$ ion will have more protons in its nucleus than the ${}^{58}_{28}Ni^{2+}$ ion.

C In the same strength magnetic field, the ${}^{60}_{28}Ni^{2+}$ ion will be deflected more than the ${}^{58}_{28}Ni^{2+}$ ion.

D These Ni^{2+} ions have the same number of electrons but a different number of neutrons.

3. The plot below shows the variation of the first nine successive ionisation energies of an element **Z**.



Which information may be inferred from the above plot?

A **Z** has no unpaired valence electrons in its atomic structure.

B **Z** has more than two occupied quantum shells in its atomic structure.

C **Z** is likely to form a covalent compound with a non-metallic element.

D **Z** is likely to achieve noble gas electronic configuration after removing its first nine electrons.

4. Which of the following statements is correct?

A The first ionisation energy increases from sodium to phosphorus as the number of protons increases but number of inner quantum shells remains the same.

B The second ionisation energy of chromium is lower than the second ionisation energy of manganese as manganese has one more proton than chromium.

C The second ionisation energy of any element is always higher than its first ionisation energy as more energy is required to remove electron from an increasingly more positive species.

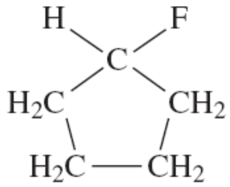
D The third ionisation energy of iron is higher than the third ionisation energy of manganese as the effective nuclear charge of iron is higher than the effective nuclear charge of manganese.

5. The following compounds all have M_r = 88. Which one contains over 60% by mass of carbon **and** also exhibits hydrogen bonding?

A $H_2N(CH_2)_4NH_2$

B $CH_3CH_2CH_2COOH$

C $CH_3CH_2CH_2CH_2CH_2OH$

D 

6. Which one of the following statements is correct?

A $AlCl_3$ is pyramidal.

B $AlCl_3$ has a higher melting point than Al_2O_3.

C The Al_2Cl_6 dimer contains two co-ordinate bonds.

D The $AlCl_3$ catalyst acts as an electron pair donor in the acylation of benzene.

7. Which of the following best accounts for the observation that gases are easily compressed?

A Gas molecules have negligible attractive forces for one another.

B The volume occupied by the gas is much greater than that occupied by the molecules.

C The average energy of the molecules in a gas is proportional to the absolute temperature of the gas.

D The collisions between gas molecules are elastic.

8. At 1.5 atm and 27°C, 1 mole of gas **A** occupies a volume of 16.7 dm^3 while 1 mole of gas **B** occupies a volume of 24 dm^3.

Which one of the following gives a possible explanation for the observation?

A The gas molecules for gases **A** and **B** experience intermolecular Van der Waals' forces of attraction.

B Gases **A** and **B** are ideal.

C Gas **A** may be He while gas **B** may be CO_2.

D Gas **A** may be NH_3 while gas **B** may be H_2.

9. An equation for the incomplete combustion of butane in oxygen is

$$C_4H_{10} + 4\tfrac{1}{2}O_2 \rightarrow 4CO + 5H_2O$$

The volume in dm^3 of oxygen at 295 K and 100 kPa required to burn 0.10 mol of butane to form steam and carbon monoxide only is

A 8.6 **B** 11 **C** 12 **D** 16

10. Consider the half-equations given below.

	$E^{\ominus}/V$
$VO_2^{+}(aq) + 2H^{+}(aq) + e^{-} \longrightarrow VO^{2+}(aq) + H_2O(l)$	+1.00
$Cr_2O_7^{2-}(aq) + 14H^{+}(aq) + 6e^{-} \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$	+1.33
$Co^{3+}(aq) + e^{-} \longrightarrow Co^{2+}(aq)$	+1.82

Which one of the following statements is **not** correct?

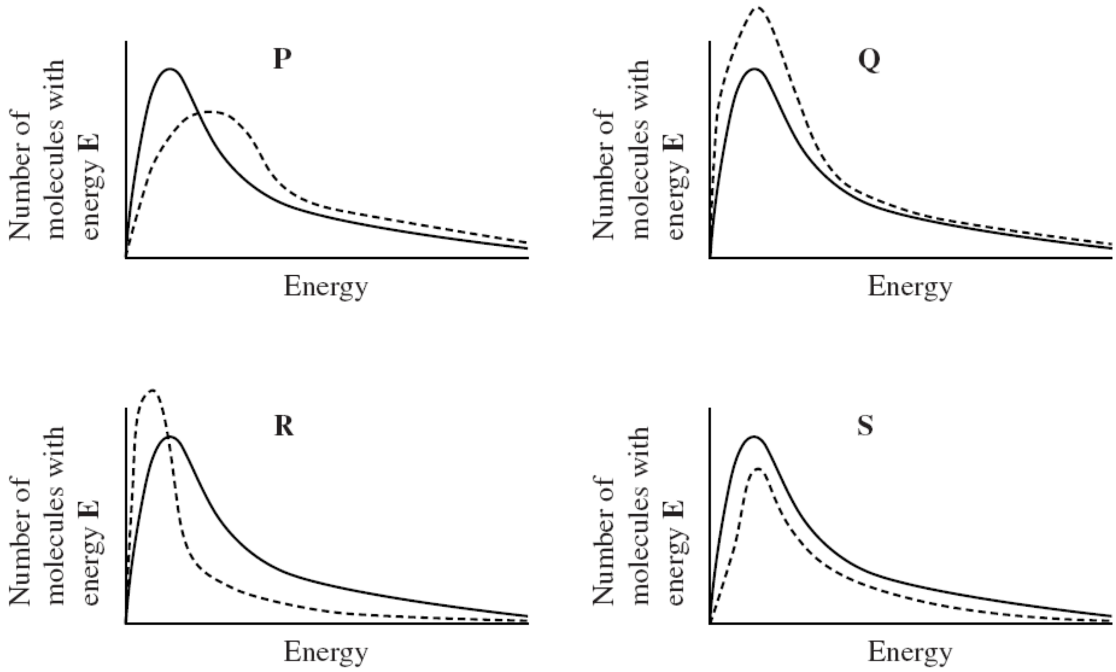
A The electron arrangement of a Co^{3+} ion is $[Ar]3d^6$.

B When $VO_2^{+}(aq)$ forms $VO^{2+}(aq)$, the oxidation state of vanadium changes from +5 to +4.

C Acidified potassium dichromate(VI) can oxidise $VO^{2+}(aq)$ to $VO_2^{+}(aq)$ under standard conditions.

D An acidified solution containing $VO_2^{+}(aq)$ ions can oxidise $Co^{2+}(aq)$ to $Co^{3+}(aq)$ under standard conditions.

11. The diagrams **P**, **Q**, **R and S** show how a change in conditions affects the Maxwell-Boltzmann distribution of molecular energies for gas G. In each case, the original distribution is shown by a solid line and the distribution after a change has been made is shown by a dashed line.



Which one of the following statements is correct?

A the change shown in diagram **Q** occurs when a catalyst is used.

B the change shown in diagram **R** occurs when the temperature is increased.

C the change shown in diagram **P** occurs when the temperature is decreased.

D the change shown in diagram **S** occurs when the pressure of G is decreased at constant temperature.

12. $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

The above reaction can be said to have reached equilibrium when

A the equilibrium constant K is equal to 1.

B the reaction between the acid and the alcohol has stopped.

C the concentrations of the products equal those of the reactants.

D the rate of production of ethyl ethanoate equals its rate of hydrolysis.

13. Public swimming pools are often chlorinated to kill bacteria. As an alternative to chlorination, silver ions can be used in a concentration of not more than 10^{-6} mol dm^{-3} and not less than 10^{-7} mol dm^{-3} of silver ions.

Which of the following compounds would, in saturated solution, provide the necessary concentration of silver ion?

	compound	***solubility product***
A	AgBr	5×10^{-13} mol^2 dm^{-6}
B	Ag*Cl*	2×10^{-10} mol^2 dm^{-6}
C	$AgIO_3$	2×10^{-8} mol^2 dm^{-6}
D	Ag_2CO_3	5×10^{-12} mol^3 dm^{-9}

14. The ability of an element to act as an oxidising agent tends to increase as

A the atom becomes larger.
B the electronegativity increases.
C the ionization energy decreases.
D the element becomes more metallic.

15. Which one of the following statements is correct?

A The solubility of Group II oxides decreases down the group.
B PCl_5 is hydrolysed by water to form a weakly acidic solution.
C Transition metals have higher densities compared to Group II metals.
D HI has a higher boiling point than HC*l* because HI has stronger hydrogen bonding.

16. Which one of the following statements about the oxides SiO_2, P_4O_{10} and SO_2 is incorrect?

A They are all acidic oxides.
B They are all covalent oxides.
C Silicon (IV) oxide is the only one having a giant lattice.
D The oxidation number of silicon, phosphorus and sulphur shows an increasing trend.

17. Which of the following is not a characteristic of magnesium, calcium and barium?

A They are powerful reducing agents.
B They are good conductors of heat and electricity.
C They react with $H_2O(\mathit{l})$ readily to give off a colourless gas.
D They have higher melting points than the corresponding alkaline metals.

18. Which of the following statements explains why the thermal stability of the hydrogen halides (HX) decreases from HF to HI?

A Acidity increases from HF to HI.
B Reactivity decreases from F_2 to I_2.
C The H–X bond gets weaker from HF to HI.
D The van der Waals' forces get stronger from HF to HI.

19. An aqueous solution containing I^- ions is treated with $AgNO_3$(aq), giving a precipitate P, which is then tested for its solubility in concentrated NH_3(aq). What is the colour of P and its solubility in NH_3(aq)?

	Colour of P	Solubility in NH_3(aq)
A	White	Slightly soluble
B	Cream	Insoluble
C	Yellow	Slightly soluble
D	Yellow	Insoluble

20. In an experiment, y moles of bromine gas is reacted completely with excess aqueous sodium hydroxide at 25 °C, the resulting solution acidified and aqueous silver nitrate added. How many grams of silver bromide will be formed?

A $\frac{188y}{3}$ **B** 188y **C** $\frac{940y}{3}$ **D** 376y

21. Which one of the following ions is likely to exist?

A TiO_4^{-} **B** TiO_4^{2-} **C** TiO_2^{2-} **D** TiO_2^{2+}

22. When drops of NH_3(aq) are added to $Cu(NO_3)_2$(aq), a pale blue preciptiate is formed. This precipitate dissolves when excess NH_3(aq) is added, forming a deep blue solution. Which process does **not** occur in this sequence?

A Dative bond formation.
B Reduction of copper(II) ions.
C Formation of a complex ion.
D Precipitation of copper(II) hydroxide.

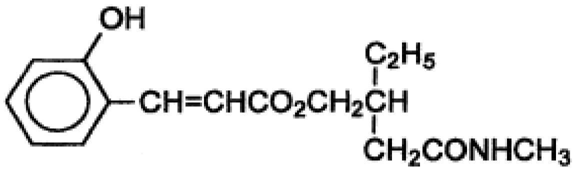
23. The compound $CF_3CHBrC\mathit{l}$ is a general anaesthetic called halothane. The number of structural isomers, including halothane, having the molecular formula $C_2HBrC\mathit{l}F_3$ is

A 2 **B** 3 **C** 4 **D** 5

24. Which one of the following sequences of nitration, alkylation and bromination is expected to give the best yield for the synthesis of 2–bromo–4–nitromethylbenzene from benzene?

A Alkylation, nitration, bromination
B Alkylation, bromination, nitration
C Nitration, alkylation, bromination
D Nitration, bromination, alkylation

25. The structure below shows the active ingredient of a certain drug.



Which one of the following statements is **FALSE** about this compound?

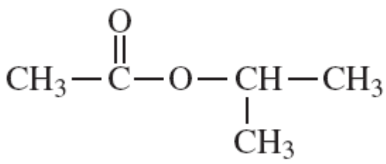
A It has 4 stereoisomers.

B It gives white fumes with PCl_5.

C One molecule of this compound reacts with 3 molecules of bromine.

D Complete hydrolysis of this compound gives CH_3NH_2 as one of the product.

26. Which one of the statements about the following ester is correct?



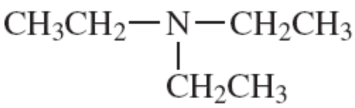
A It is a chain isomer of pentanoic acid.

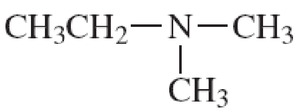
B It is a functional group isomer of ethyl propanoate.

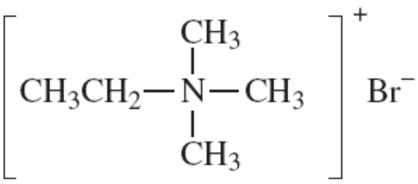
C It can be hydrolysed to produce an alcohol that is resistant to oxidation by acidified potassium dichromate(VI).

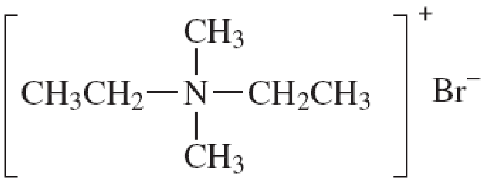
D It can be hydrolysed to produce an alcohol that can also be formed by the acid-catalysed hydration of propene.

27. Which one of the following is formed when an excess of bromomethane reacts with diethylamine?

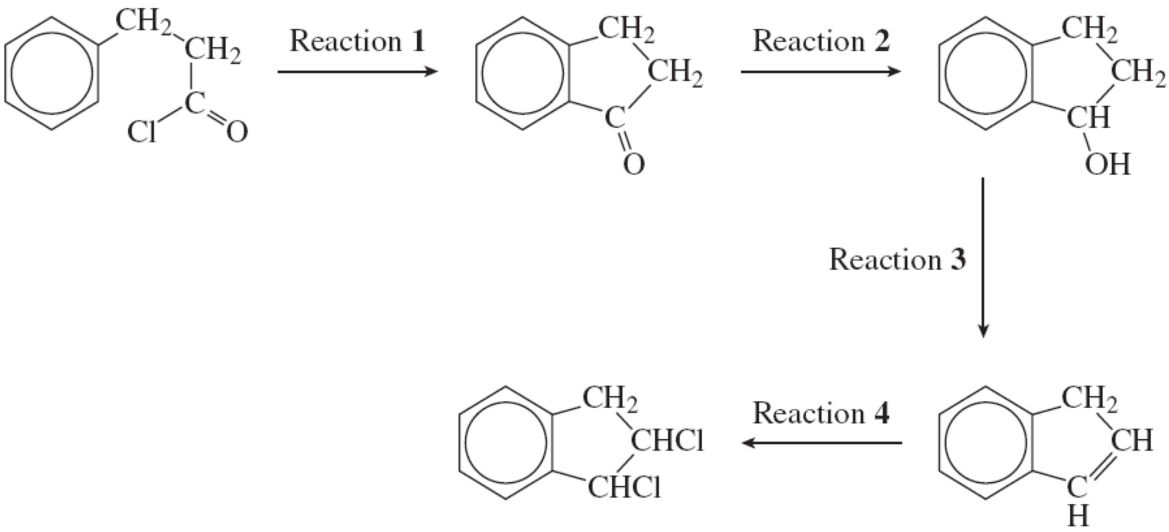
A 

B 

C 

D 

28. A sequence of reactions is shown below.

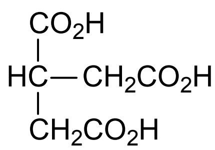


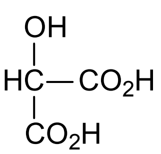
A correct list of substances for the sequence is

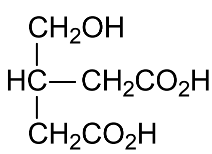
	Reaction **1**	Reaction **2**	Reaction **3**	Reaction **4**
A	CH_3COCl	$NaBH_4$	Conc. H_2SO_4	Cl_2
B	$AlCl_3$	HCl	$NaOH$	Cl_2
C	$AlCl_3$	$NaBH_4$	Conc. H_2SO_4	HCl
D	$AlCl_3$	$NaBH_4$	Conc. H_2SO_4	Cl_2

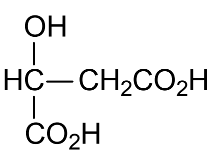
29. One industrial preparation of ethanoic acid is direct carbonylation of methanol, using a rhodium catalyst.

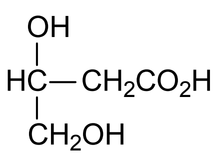
$$CH_3OH + CO \xrightarrow{rhodium} CH_3CO_2H$$

Which compound could be expected to produce  by this method?

A 

B 

C 

D 

30. Gastrins are hexadecapeptide (16 amino acid units) hormones that stimulate the secretion of gastric acid in the stomach of mammals. On hydrolysis of gastrins, the following four peptide fragments were isolated.

H_2N–gly–trp–CO_2H
H_2N–leu–glu–glu–glu–ala–ala–try–CO_2H
H_2N–glu–gly–pro–trp–CO_2H
H_2N–met–asp–phe–CO_2H

C–terminus analysis of the hexadecapeptide yields a *phe* residue while N–terminus analysis yields a *glu* residue.

Based on the information above, suggest a primary structure for the hexadecapeptide and the number of peptide bonds it contains.

	Primary structure	No. of peptide bonds
A	*H_2N–glu-gly-pro-trp-trp-gly-leu-glu-glu-glu-ala-ala-try-met-asp-phe–CO_2H*	15
B	*H_2N–glu-gly-pro-trp-gly-trp-leu-glu-glu-glu-ala-ala-try-met-asp-phe–CO_2H*	15
C	*H_2N–glu-gly-pro-trp-gly-trp-leu-glu-glu-glu-ala-ala-try-met-asp-phe–CO_2H*	16
D	*H_2N–phe-asp-met-try-ala-ala-glu-glu-glu-leu-trp-gly-trp-pro-gly-glu–CO_2H*	16

SECTION B (10 MARKS)

*For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct. The responses **A** to **D** should be selected on the basis of*

A	**B**	**C**	**D**
1, 2 and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is to be used as a correct response

31. The extraction of titanium from titanium(IV) oxide involves two reactions represented by the following equations

$$TiO_2 + 2C + 2Cl_2 \rightarrow TiCl_4 + 2CO$$
$$TiCl_4 + 4Na \rightarrow Ti + 4NaCl$$

Correct statements about the extraction include

1 149.6 kg of chlorine are needed to make 200.0 kg of titanium(IV) chloride (M_r = 189.9).

2 both of the above equations represent redox reactions.

3 the second reaction is carried out in an atmosphere of nitrogen to prevent oxidation of the reactant.

32. Gas X reacts with gas Y in a container of 1 dm^3 according to the following equation:

$$X(g) + Y(g) \rightarrow XY(g)$$

The rate equation for the reaction is rate = $k[X][Y]^2$

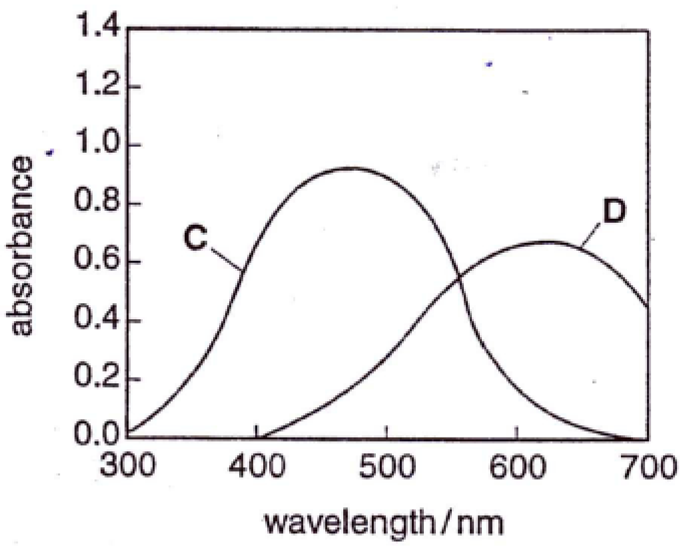
At constant temperature, which of the following statement(s) is/are correct?

1 units of k is $mol^{-2}\ dm^6\ s^{-1}$.

2 halving the volume of the container will decrease the rate by a factor of 8.

3 quadrupling the concentration of Y, keeping the concentration of X constant, will increase the rate by a factor of 64.

33. The visible spectra of solutions of two transition metal complexes **C** and **D** are shown in the diagram below. Both complexes contain the same transition metal ion.



Colour	Wavelength / nm
Violet	390 – 455
Blue	455 – 492
Green	492 – 577
Yellow	577 – 597
Orange	597 – 622
Red	622 – 780

Which of following statements can be deduced from the spectra?

1 Complex **C** is likely to be red while complex **D** is likely to be violet.

2 The ligand in complex **C** is a stronger field ligand than that in complex **D**

3 The K_c value for the formation of complex **C** is higher than the K_c value for formation of complex **D**.

34. Which of the following statement(s) is/are correct?

1 the melting point increases from pentan-3-one to pentan-2-ol to 2-aminopropanoic acid.

2 the base strength increases from methylamine to ammonia to phenylamine.

3 the carbon to carbon bond energy increases from ethene to benzene to ethane.

35. Solutions that form bubbles of a gas with solid Na_2CO_3 include

1 CH_3CHO(aq) **2** HCOOH(aq) **3** $CrCl_3$(aq)

36. A substitution reaction occurs when ammonia reacts with

1 CH_3Br **2** $[Cu(H_2O)_6]^{2+}$ **3** HBr

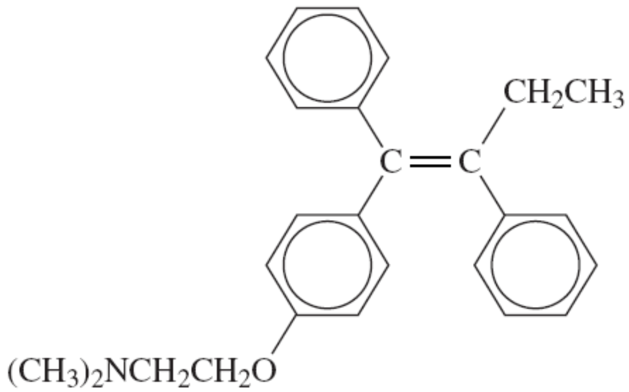
37. Reactants that form an organic product which has an asymmetric carbon atom include

1 but-2-ene and HBr
2 propanal and HCN
3 propanone and $NaBH_4$

38. A car burning lead-free fuel has a catalytic converter fitted to its exhaust. On analysis, its exhaust gases are shown to contain small quantities of nitrogen oxides. Which modifications would result in lower exhaust concentrations of nitrogen oxides?

1 an increase in the surface area of the catalyst in the converter.
2 an increase in the rate of flow of the exhaust gases through the converter.
3 a much higher temperature of combustion in the engine.

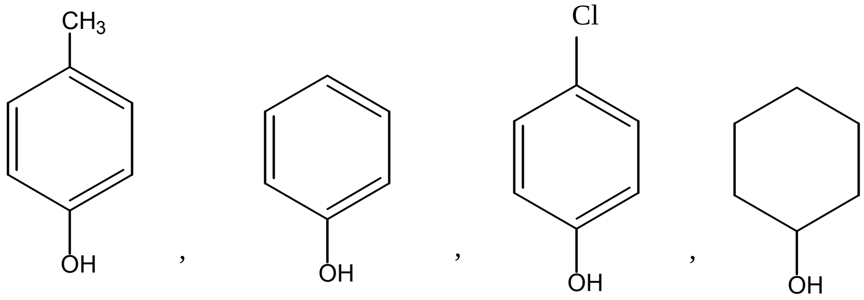
39. The drug tamoxifen, which is used in the treatment of cancer, has the structure

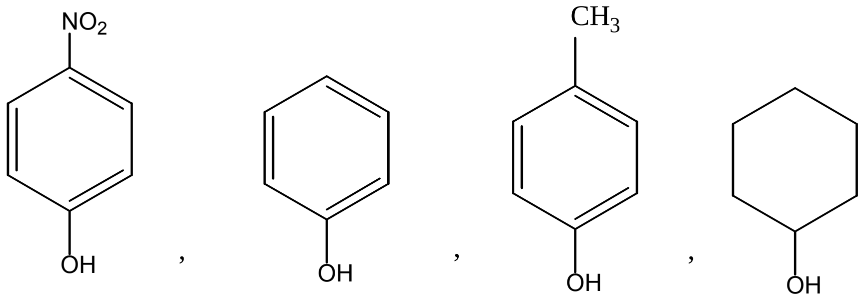


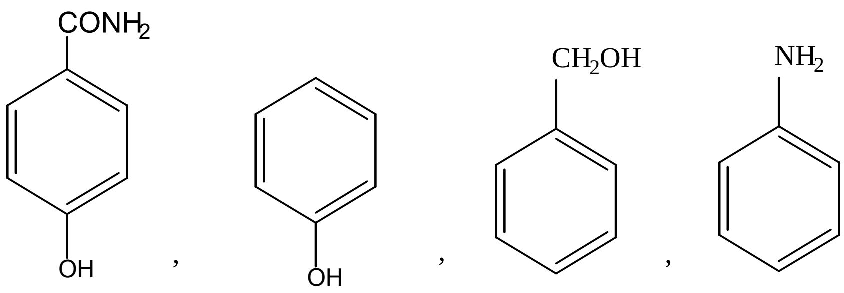
Which of the following statement(s) about tamoxifen is/are correct?

1 it is insoluble in HC*l*(aq).
2 it has a stereoisomer.
3 it can undergo free radical substitution with chlorine.

40. Which of the following correctly list(s) the compounds in order of decreasing acidity?

1 

2 

3 

~ All the Best! ~